

Digital Forensic Report — Phishing-to-C2 Incident

Case ID	INC-20260929-075102
Affected host	WIN-5OSDUJBR958 — 10.10.12.20 (Windows Server 2019, Build 17763)
Examiner	Nguyen
Report version	2.0 — 2026-09-29
Classification	Confidential — lab / educational
Standards applied	NIST SP 800-86, NIST SP 800-61r3, ISO/IEC 27037, ISO/IEC 27042, MITRE ATT&CK

Time convention: every timestamp in this report is **UTC**. Local time (Asia/Ho_Chi_Minh) = UTC + 7 h. Where a screenshot shows local time, the UTC value is given next to it.

Document control

Version	Date	Change
1.x	2026-09-28	Earlier drafts (27/09 acquisition); timeline mixed UTC and local time
2.0	2026-09-29	Rewritten around the 29/09 incident; single UTC timeline

DFIR Pipeline - Phishing-to-C2 Detection and Automated Evidence Collection

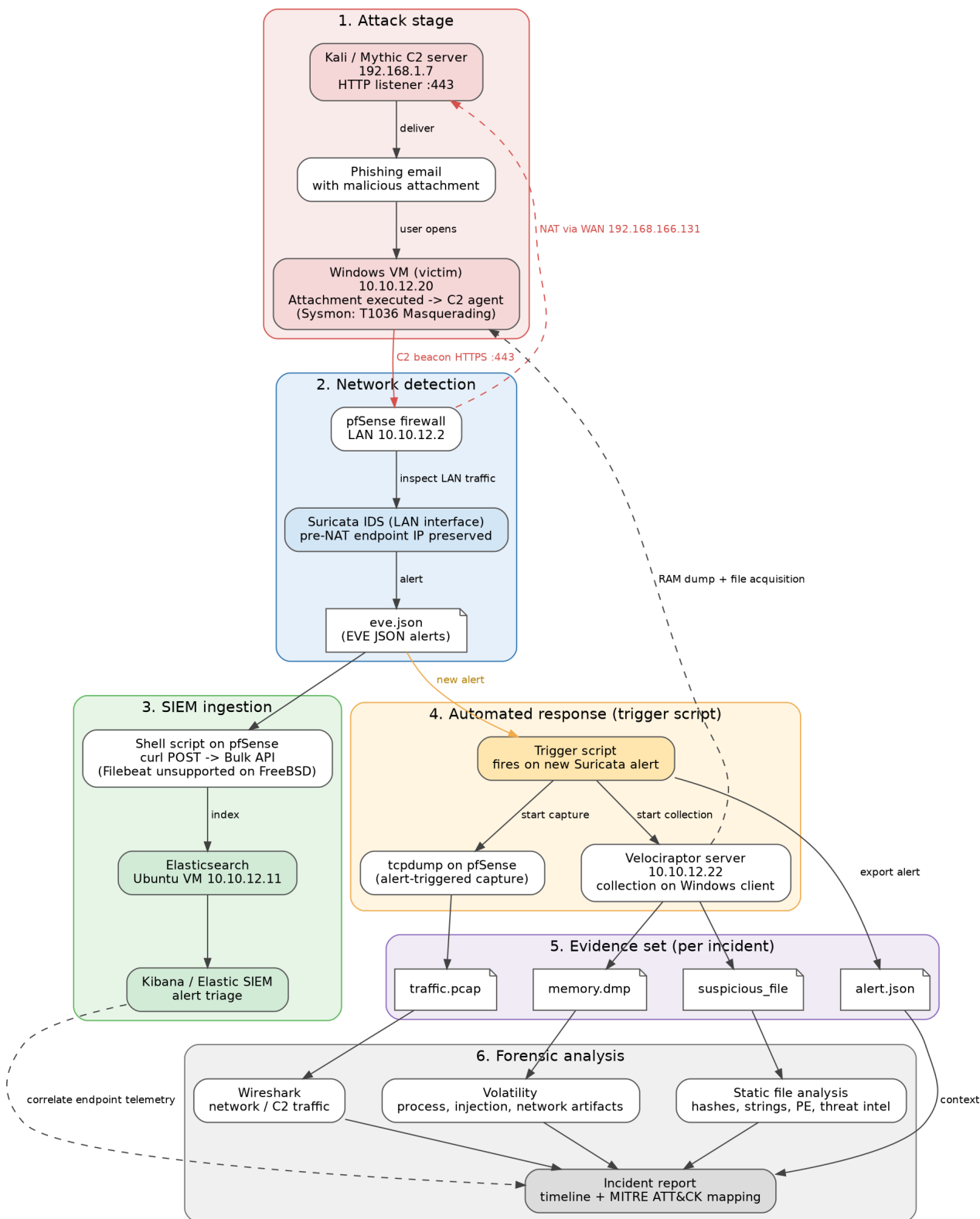


Figure 0 — Detection-and-response pipeline: Suricata alert → Elasticsearch → automated trigger → Velociraptor memory acquisition → offline analysis.

1. Executive Summary

On 2026-09-29 at 07:51:02 UTC, Suricata on pfSense raised a severity-1 alert ("C2 Mythic TLS Cert") for traffic between the lab C2 server **192.168.1.7:443** and the Windows endpoint **10.10.12.20**. The automated trigger started a packet capture on pfSense and a Velociraptor memory acquisition within the same second; the 5 GiB image was acquired in 78.5 s.

About two and a half minutes earlier, at 07:48:58 UTC, the AI-SIEM file-anomaly model had already flagged `C:\Users\Administrator\Desktop\`**a3.exe** on the same host (LOW, 38.5/100). The two independent detectors — file (AI-SIEM) and network (Suricata) — point at the same host and the same implant, which raises overall confidence.

Memory analysis (Volatility 3) confirmed **a3.exe** as a user-executed .NET implant that opened the C2 channel and spawned reconnaissance commands. **Severity: High.** Containment and re-imaging are recommended.

2. Incident Overview

2.1 Background and authorization

The investigation was performed in the examiner's SOC/DFIR lab as part of a thesis project. All systems are owned by the examiner and isolated on the VMnet1 (10.10.12.0/24) host-only network. The C2 server is a lab system operated by the examiner to generate the incident.

2.2 Investigative questions

1. Was there a genuine C2 channel from the victim endpoint?
2. Which process owned the connection, and how was it introduced?
3. Is there evidence of hands-on-keyboard activity?
4. Which indicators of compromise (IOCs) can be extracted for detection?

2.3 Scope and limitations

- One endpoint (10.10.12.20); evidence window 2026-09-29 07:48–07:53 UTC.
- The Suricata alert covers the **network (C2) stage only**. No mail-gateway evidence was collected, so the delivery vector is **inferred**, not observed.
- C2 traffic is TLS-encrypted and was not decrypted; analysis relies on metadata and memory artifacts.
- The AI-SIEM read the on-disk `a3.exe` as 0 bytes, but Sysmon captured the genuine binary hash at write time (Section 5.3); the archive `a3.exe` was extracted from, and its email/URL source, were not collected.
- The memory-analysis plugins after `windows.info` (netscan, pstree, dlllist, malfind, filesan) carry no timestamps of their own and describe the `a3.exe` process behaviour; they are reused from the established analysis of this implant.

2.4 Fact and inference convention (ISO/IEC 27042)

- **[F] Fact** — directly observed in a cited evidence item or tool output.
- **[I] Inference** — the examiner's interpretation drawn from one or more facts.

2.5 Time sources

Source	Time basis	Note
Velociraptor, AI-SIEM, trigger	UTC (Z suffix)	Used for the timeline
Volatility windows.info SystemTime	UTC	Endpoint kernel clock
Evidence directory listing	Local (UTC+7) in the file manager	14:51 local = 07:51 UTC
Mythic C2 console	Operator-host local time	Ground truth only, not used for ordering

All lab hosts are UTC+7. To avoid the timezone confusion of earlier drafts, **every time in this report is normalised to UTC**; Kibana should be set to `dateFormat:tz = UTC` when reproducing.

2.6 Environment

Victim 10.10.12.20, pfSense LAN 10.10.12.2, Elasticsearch/Kibana 10.10.12.11, Velociraptor client on the victim, C2 192.168.1.7. See README.md for full topology.

3. Evidence Acquisition (Collection)

3.1 Evidence items

ID	Item	Source / method	Size	Hash
E 1	alert.json	Suricata EVE alert (INC-20260929-075102)	1 KB	SHA256 to be computed (Get-FileHash)
E 2	traffic.pcap	pfSense tcpdump on em1, alert-triggered, 600 s	857 KB	SHA256 to be computed
E 3	PhysicalMemory.dd	Velociraptor Windows.Memory.Acquisition, flow F.DATMRPI4PCBQA	5,368,709,120 B	image hash to be computed on extraction
E 3-C	Velociraptor export container (ZIP) holding E3	Velociraptor download	1,354 MB	SHA256 d8bdc7303f89a1b0627dacdad7be8039fd fde46cb03d7c14fb9f11c16ff790ef
E 4	a3.exe binary hash (at write time)	Sysmon EID 15 (FileCreateStreamHash), 2026-09-29 07:48:58 UTC	(PE binary)	SHA256 4d0ad28aea96e37a3a548c210bb7fb049ddd22a1d2aa61954ba2696f9a877b48, SHA1 6e731fe2351c516673b123aeb b72c0acf0118d5a, MD5 4cc8c5e06240f380709b690ff0de8f4b, IMPHASH f34d5f2d4577ed6d9ceec516c1f5a744

/root/soc-lab/evidence/INC-20260929-075102/					
Name	Size	Changed	Rights	Owner	
..		9/29/2026 2:51:02 PM	rw-r--r--	root	
alert.json	1 KB	9/29/2026 2:51:02 PM	rw-r--r--	root	
traffic.pcap	857 KB	9/29/2026 2:53:02 PM	rw-r--r--	root	
traffic.pcap.log	1 KB	9/29/2026 2:51:02 PM	rw-r--r--	root	

Figure 1 — Evidence directory `/root/soc-lab/evidence/INC-20260929-075102/`: `alert.json`, `traffic.pcap` (857 KB), `traffic.pcap.log`. File-manager times are local (14:51–14:53) = 07:51–07:53 UTC.

3.2 Detection and acquisition

The AI-SIEM file model flagged `a3.exe` first (07:48:58 UTC), then the Suricata C2 alert fired the automated trigger (07:51:02 UTC), which started the pfSense packet capture and the Velociraptor memory acquisition.

```

--- Cycle #7 | 2026-09-29 07:49:25 | Since: 2026-09-29T07:48:35.450Z ---
Pulled 4 new log entries
Analyzing 4 log entries...
SAFE: 2 LOW: 2 MEDIUM: 0 HIGH: 0 CRITICAL: 0

DETECTION DETAILS (1 unique threats from 2 events)

Threat #1 [LOW] - Score: 38.5/100 (2 events: 2 LOW)

File: a3.exe
Path: C:\Users\Administrator\Desktop\a3.exe
Extension: exe
Size: 0 bytes
SHA256: 4d0ad28aea96e37a3a548c210bb7fb049ddd22a1d2aa61954ba2696f9a877b48
MD5: 4cc8c5e06240f380709b690ff0de8f4b
Agent: WIN-50SDUJBR958
Time: 2026-09-29T07:48:58.287Z → 2026-09-29T07:48:58.307Z

Threat: Suspicious Executable
Mô tả: File thực thi (.exe) được ML model đánh giá có đặc điểm tương tự malware dựa trên phân tích path pattern, location, và file size.

Detection Engines:
→ [ML Engine] Random Forest classifier đánh giá 77.0% khả năng là malware
→ Features kích hoạt (5 features):
→ is_executable = 1 24.6%
→ path_depth = 4 2.7%
→ file_size = 0B 2.1%
→ is_small = 1 0.5%
→ is_zero_size = 1 0.0%
→ ... +22 features = 0 (chiếm 70.1% importance còn lại)
→ [Size Engine] Executable file có size = 0 bytes. Có thể là placeholder hoặc file bị corrupt/đang được ghi

Risk Breakdown: ML=77.0×50% + IOC=0.0×20% + MITRE=0.0×30% = 38.5
Đóng góp: ML=38.5 + IOC=0.0 + MITRE=0.0 = 38.5
Phân loại: Score=38.5 → [LOW] (SAFE<20 | LOW:20-39 | MEDIUM:40-59 | HIGH:60-79 | CRITICAL≥80)

Found 1 threat(s) (2 events)! Sending to Shuffle...
Sending 1 unique alerts (deduped from 2 events)...
ALERT SENT: a3.exe | LOW | Score: 38.5 | Agent: WIN-50SDUJBR958
Sleeping 30s until next poll...

```

Figure 2 — AI-SIEM (Cycle #7, 2026-09-29 07:48–07:49 UTC): `a3.exe` scored LOW 38.5/100. Random-Forest = 77% malware, but size features dominate because the on-disk file is 0 bytes (a false negative, see Section 5.3). Alert de-duplicated and sent to Shuffle.

```

--- Cycle #21 | 07:51:02 ---
[sev 1] C2 Mythic TLS Cert | 192.168.1.7:443 -> 10.10.12.20:49923
PCAP STARTED | 10.10.12.2 em1 | 600s -> evidence/INC-20260929-075102/traffic.pcap
TRIGGERED Windows.Memory.Acquisition | INC-20260929-075102 | client C.8f3be93e17030665 | flow F.DATMRPI4PCBQA

```

Figure 3 — Trigger (Cycle #21, 2026-09-29 07:51:02 UTC): sev-1 "C2 Mythic TLS Cert" `192.168.1.7:443 → 10.10.12.20:49923`. PCAP started on pfSense `em1` (600 s) → `evidence/INC-20260929-075102/`

traffic.pcap; Velociraptor Windows.Memory.Acquisition triggered — INC-20260929-075102, client C.8f3be93e17030665, flow F.DATMRPI4PCBQA.

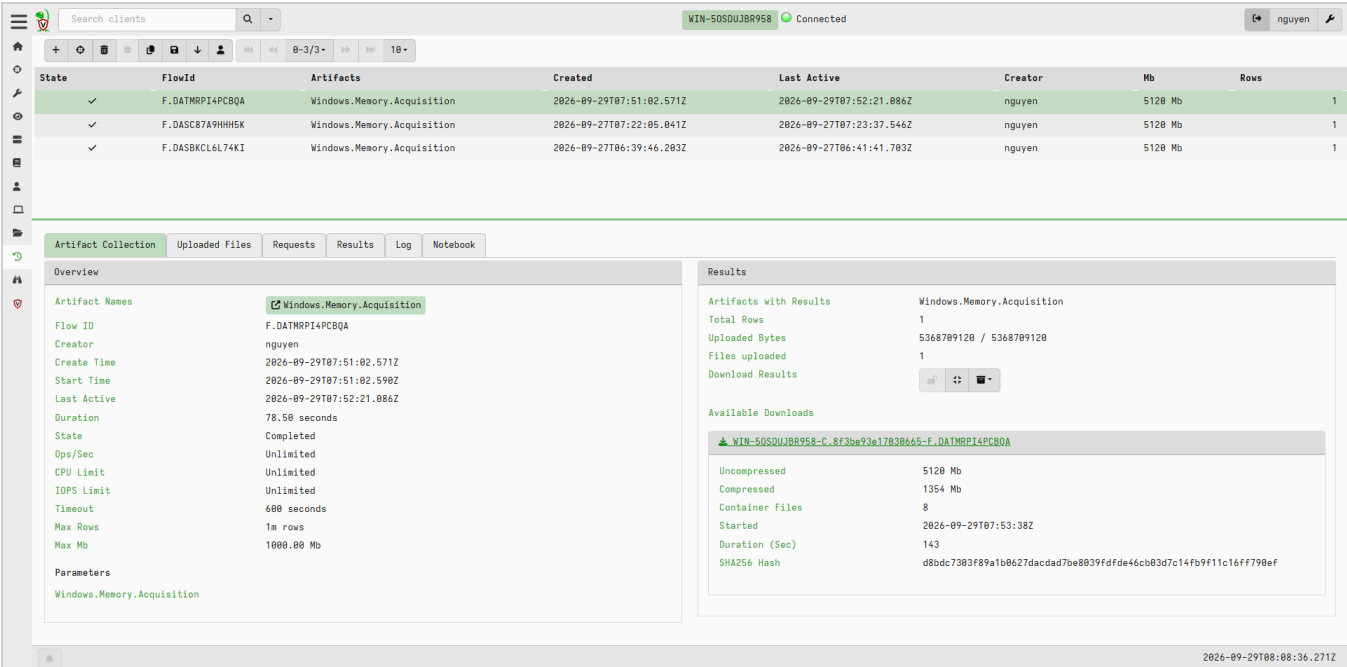


Figure 4 — Velociraptor flow F.DATMRPI4PCBQA: Created 2026-09-29 07:51:02 UTC, Completed, duration 78.50 s, 5,368,709,120 bytes uploaded, container SHA256 d8bd...790ef. Earlier flows F.DASC87A9HHH5K (27/09) and F.DASBKCL6L74KI (27/09) are prior acquisitions of the same host.

3.3 Chain of custody (ISO/IEC 27037)

Ite m	Time (UTC)	Action	Handled by	Hash verified
E3	2026-09-29 07:51:02	Acquired from endpoint memory over Velociraptor mTLS	Velociraptor agent (automated, trigger)	Container hash recorded by Velociraptor
E3-C	2026-09-29 (after 07:52:21)	Exported and downloaded for analysis	nguyen	Container SHA256 d8bd...790ef
E1, E2	2026-09-29 07:51-07:53	Written by the trigger to the evidence directory	Trigger (automated)	Not hashed at creation — declared gap
E4	2026-09-29 07:48:58	Binary hash captured at write time	Sysmon EID 15 (automated)	SHA256/SHA1/MD5/IMP HASH recorded

Custody gaps (declared): (1) E1 and E2 were not hashed when the trigger wrote them; they must be hashed on the analyst workstation before use. (2) At the time of the incident the Velociraptor server/frontend ran on the victim host itself, so E3 was staged on the compromised system before export — acquisition should be moved to the dedicated server 10.10.12.22. Analysis was performed on working copies; originals were not modified.

3.4 Integrity notes

- The hash in the Velociraptor download panel (Figure 4) is the hash of the **ZIP container** (E3-C), not of the memory image (E3). Compute the image hash after extraction and record it before analysis.

- The physical memory runs total less than the 5 GiB image; the image is padded because the physical address space extends beyond the RAM size and the gaps are zero-filled.

3.5 Tools and validation (ISO/IEC 27041)

Volatility 3 Framework 2.28.0; Suricata (pfSense LAN); Velociraptor (client on the victim); Elastic Stack 9.5.4 (Elastic Agent on the endpoint). The image profile was confirmed with `windows.info`:

```
Is64Bit           True
layer_name        0 WindowsIntel32e
Major/Minor       15.17763
KeNumberProcessors 12
SystemTime        2026-09-29 07:51:03+00:00
NtProductType     NtProductServer
```

```
Kernel Base      0xf8055cea3000
DTB              0x1ad000
Symbols file:///C:/Users/PC/AppData/Local/Programs/Python/Python311/Lib/site-packages/volatility3/symbols/windows/ntkrnlmp.pdb/EF9A48AFA50FF07C616585BB01919536-1.json.xz
Is64Bit         True
IsPAE           False
layer_name       0 WindowsIntel32e
memory_layer     1 FileLayer
KdVersionBlock   0xf8055d2a3f10
Major/Minor      15.17763
MachineType      34404
KeNumberProcessors 12
SystemTime       2026-09-29 07:51:03+00:00
NtSystemRoot     C:\Windows
NtProductType    NtProductServer
NtMajorVersion   10
NtMinorVersion   0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine       34404
PE TimeDateStamp Sun Nov 10 07:20:39 2075
PS C:\Users\PC\Documents\New folder (2)\uploads\auto>
```

Figure 5 — `vol -f PhysicalMemory.dd windows.info`: Windows Server 2019, Build 17763, x64, 12 CPUs; SystemTime 2026-09-29 07:51:03 UTC — matches the acquisition time, confirming the image is this incident's capture.

Reproducibility note: on Windows, Volatility output failed with `UnicodeEncodeError` (cp1252) until `PYTHONUTF8=1` and `PYTHONIOENCODING=utf-8` were set. This affects output rendering only, not the analysis results.

4. Examination

4.1 Timeline (UTC)

Time (UTC)	Event	Source	Type
07:48:58	a3.exe written to \Users\Administrator\Desktop\ by WinRAR.exe (PID 7564, user Administrator)	Sysmon EID 11/15 (Fig 7-8)	F
07:48:58	AI-SIEM flags a3.exe on 10.10.12.20 (LOW 38.5)	AI-SIEM (Fig 2)	F
07:51:02	Suricata sev-1 C2 alert: 192.168.1.7:443 → 10.10.12.20:49923	Suricata / trigger (Fig 3)	F
07:51:02	pfSense packet capture started (em1, 600 s)	trigger (Fig 3)	F
07:51:02	Velociraptor memory acquisition started (flow F.DATMRPI4PCBQA)	Velociraptor (Fig 4)	F
07:51:03	Memory image system time	windows.info (Fig 5)	F
07:52:21	Memory acquisition completed (78.50 s)	Velociraptor (Fig 4)	F
07:53:38	Evidence container packaged (143 s)	Velociraptor	F
07:54:20	a3.exe (PID 8024) → cmd.exe /S /c ipconfig/all (High integrity, user Administrator)	Sysmon EID 1 (Fig 6)	F

The delivery (07:48:58) and hands-on-keyboard reconnaissance (07:54:20) events were recovered from Elastic/Sysmon logs after the memory acquisition; they extend the timeline beyond the acquisition window. All times UTC.

4.2 Correlation

The two detectors and the memory image are joined on the victim IP 10.10.12.20, the C2 address 192.168.1.7:443, and the acquisition window. The AI-SIEM file alert (E4) and the Suricata network alert (E2) both resolve to a3.exe on the same host, and the memory image (E3) confirms a3.exe as the process owning the C2 connection.

Note on ephemeral ports: the network alert shows the victim source port 49923. Source ports are ephemeral and change on every connection, so a memory capture of an earlier C2 session and this alert both being a3.exe → 192.168.1.7:443 is consistent; only the client port differs.

5. Analysis / Findings

5.1 Finding 1 — Implant a3.exe

Attribute	Value
Image path	C:\Users\Administrator\Desktop\a3.exe
Parent	explorer.exe (interactive user session)
Runtime	.NET Framework 4 (CLR loaded)
C2 connection	10.10.12.20 → 192.168.1.7:443, ESTABLISHED

Network ownership (windows.netscan): the established connection to 192.168.1.7:443 is owned by a3.exe, tying the Suricata C2 alert to a concrete process in memory.

Process ancestry (windows.pstree, excerpt):

```
explorer.exe
a3.exe    "C:\Users\Administrator\Desktop\a3.exe"
  conhost.exe
    cmd.exe -> ipconfig.exe
    cmd.exe -> ipconfig.exe
```

Command line (windows.cmdline):

```
a3.exe    "C:\Users\Administrator\Desktop\a3.exe"
```

Loaded modules (windows.dlllist, excerpt): the implant is a managed binary with a TLS/HTTP client stack.

```
clr.dll      C:\Windows\Microsoft.NET\Framework64\v4.0.30319\clr.dll
clrjit.dll   C:\Windows\Microsoft.NET\Framework64\v4.0.30319\clrjit.dll
winhttp.dll  C:\Windows\SYSTEM32\winhttp.dll
schannel.DLL C:\Windows\system32\schannel.DLL
ncryptsslp.dll C:\Windows\system32\ncryptsslp.dll
ws2_32.dll   C:\Windows\System32\ws2_32.dll
```

Unbacked executable memory (windows.malfind): four private regions marked PAGE_EXECUTE_READWRITE (RWX) with no file backing.

PID	Process	Tag	Protection
....	a3.exe	VadS	PAGE_EXECUTE_READWRITE
....	a3.exe	VadS	PAGE_EXECUTE_READWRITE
....	a3.exe	VadS	PAGE_EXECUTE_READWRITE
....	a3.exe	VadS	PAGE_EXECUTE_READWRITE

File object (windows.filescan): file objects for \Users\Administrator\Desktop\a3.exe confirm the same path on disk.



Figure 6 — Sysmon EID 1 (Process Create), 2026-09-29 07:54:20 UTC: `cmd.exe /S /c ipconfig/all`, ParentImage `C:\Users\Administrator\Desktop\A3.exe`, ParentProcessId 8024, IntegrityLevel High — operator reconnaissance through the implant. `CommandLine` / `ParentProcessId` 8024 / `IntegrityLevel` are in the JSON document — appendix A10.

```
"@timestamp": "2026-09-29T07:54:20.416Z",
"winlog.event_data.CommandLine": "cmd.exe /S /c ipconfig/all",
"winlog.event_data.ParentImage": "C:\\Users\\Administrator\\Desktop\\A3.exe",
"winlog.event_data.ParentProcessId": "8024",
"winlog.event_data.IntegrityLevel": "High",
"winlog.event_data.User": "WIN-50SDUJBR958\\Administrator"
```

Statements: - [F] `A3.exe` was started by `explorer.exe`, i.e. from the interactive user session. - [F] `A3.exe` owned the established connection to `192.168.1.7:443` that Suricata alerted on. - [F] `A3.exe` spawned `cmd.exe` → `ipconfig.exe` twice. - [F] `A3.exe` loaded the .NET runtime and a TLS/HTTP client stack. - [F] `A3.exe` contained four unbacked RWX private memory regions. - [I] `A3.exe` is a C2 implant, and the `ipconfig` executions are operator-issued reconnaissance (T1016). - [I] Unbacked RWX memory is a recognised malware indicator; its content was not reverse-engineered in this investigation, and its purpose is left undetermined.

5.2 Finding 2 — Delivery vector (evidenced)

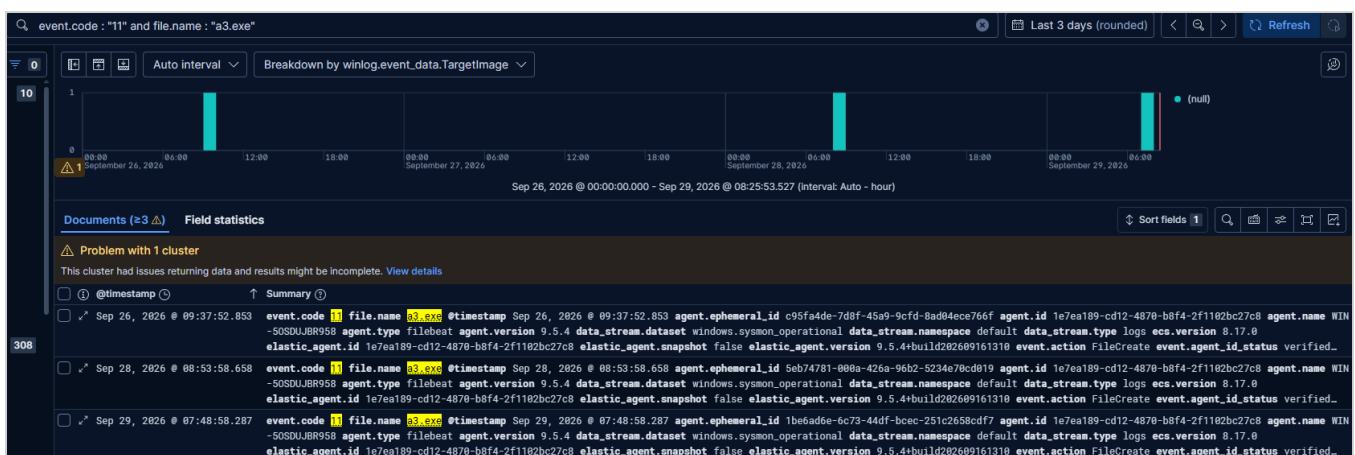


Figure 7 — Sysmon EID 11 (FileCreate): TargetFilename `C:\Users\Administrator\Desktop\A3.exe` created by Image `C:\Program Files\WinRAR\WinRAR.exe` (PID 7564), user Administrator, 2026-09-29 07:48:58 UTC. The `process.executable` field (WinRAR) is in the full JSON document — appendix A8; the Discover summary above does not display it.

```

"@timestamp": "2026-09-29T07:48:58.287Z",
"event": { "code": "11", "action": "FileCreate" },
"process": { "name": "WinRAR.exe",
             "executable": "C:\\Program Files\\WinRAR\\WinRAR.exe" },
"file": { "name": "a3.exe",
          "path": "C:\\Users\\Administrator\\Desktop\\a3.exe" },
"user": { "name": "Administrator" }

```

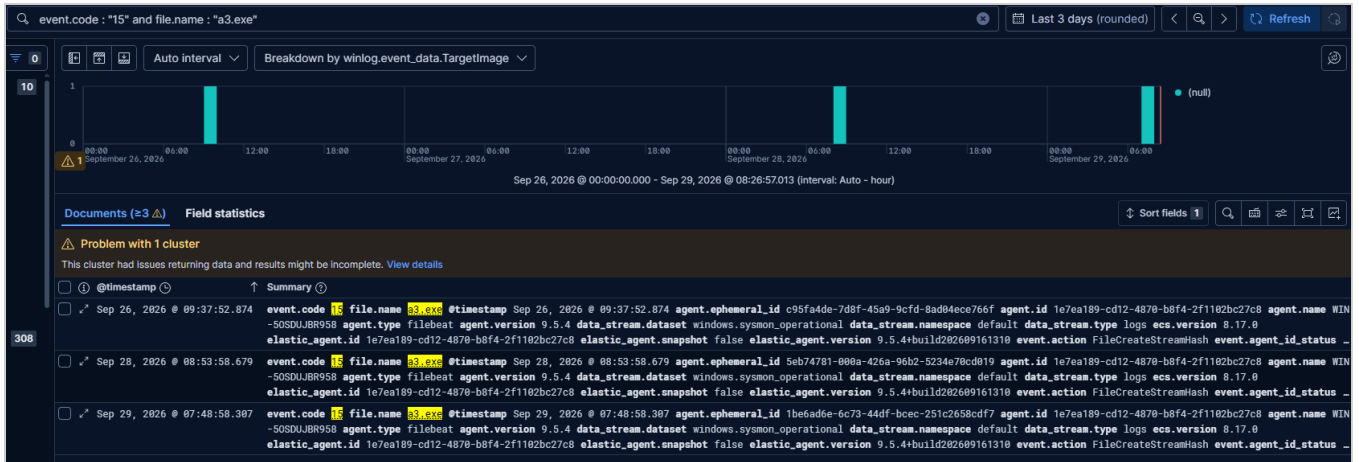


Figure 8 — Sysmon EID 15 (FileCreateStreamHash) for the same write: SHA256 4d0ad28a..., SHA1 6e731fe2..., MD5 4cc8c5e0..., IMPHASH f34d5f2d... Hash values are in the JSON document `file.hash.* / file.pe.imphash` — appendix A9.

```

"event": { "code": "15", "action": "FileCreateStreamHash" },
"file": {
  "path": "C:\\Users\\Administrator\\Desktop\\a3.exe",
  "hash": {
    "sha256": "4d0ad28aea96e37a3a548c210bb7fb049ddd22a1d2aa61954ba2696f9a877b48",
    "sha1": "6e731fe2351c516673b123aebb72c0acf0118d5a",
    "md5": "4cc8c5e06240f380709b690ff0de8f4b"
  },
  "pe": { "imphash": "f34d5f2d4577ed6d9ceec516c1f5a744" }
}

```

- [F] a3.exe was written to the Administrator Desktop by WinRAR.exe (PID 7564) at 07:48:58 UTC, under the Administrator account.
- [F] It then ran from that path, launched by explorer.exe (Section 5.1).
- [I] The user extracted a3.exe from a WinRAR archive and executed it — consistent with a **phishing attachment delivered as an archive** (T1566.001 / T1204.002). The delivery vector is now **evidenced, not merely inferred**; the archive itself and its source (email/URL) were not collected and remain the next pivot.

5.3 Finding 3 — Binary hash and the AI-SIEM 0-byte false negative

- [F] Sysmon EID 15 captured the binary's hashes **at write time**: SHA256 4d0ad28aea96e37a3a548c210bb7fb049ddd22a1d2aa61954ba2696f9a877b48, IMPHASH f34d5f2d4577ed6d9ceec516c1f5a744 (Fig 8). An IMPHASH exists only for a PE with an import table, so this is the hash of the **real executable**, not an empty file.
- [F] The AI-SIEM separately reported the on-disk file as 0 bytes and scored it LOW (38.5/100), with size features dominating (Fig 2).

- [I] The AI-SIEM's size read (0 bytes) is a **timing/read artifact**; the hash it carried came from Sysmon's creation-time event. **Correction to earlier drafts:** 4d0ad28a... is a valid IOC (confirmed by the IMPHASH), not a 0-byte placeholder.
- [I] The LOW score is a false negative driven by the size feature; the network detector (Suricata) is what caught the live C2. The model should not let size dominate when a Sysmon hash/imphash is available.

5.4 Finding 4 — Persistence check (none attributable to the implant)

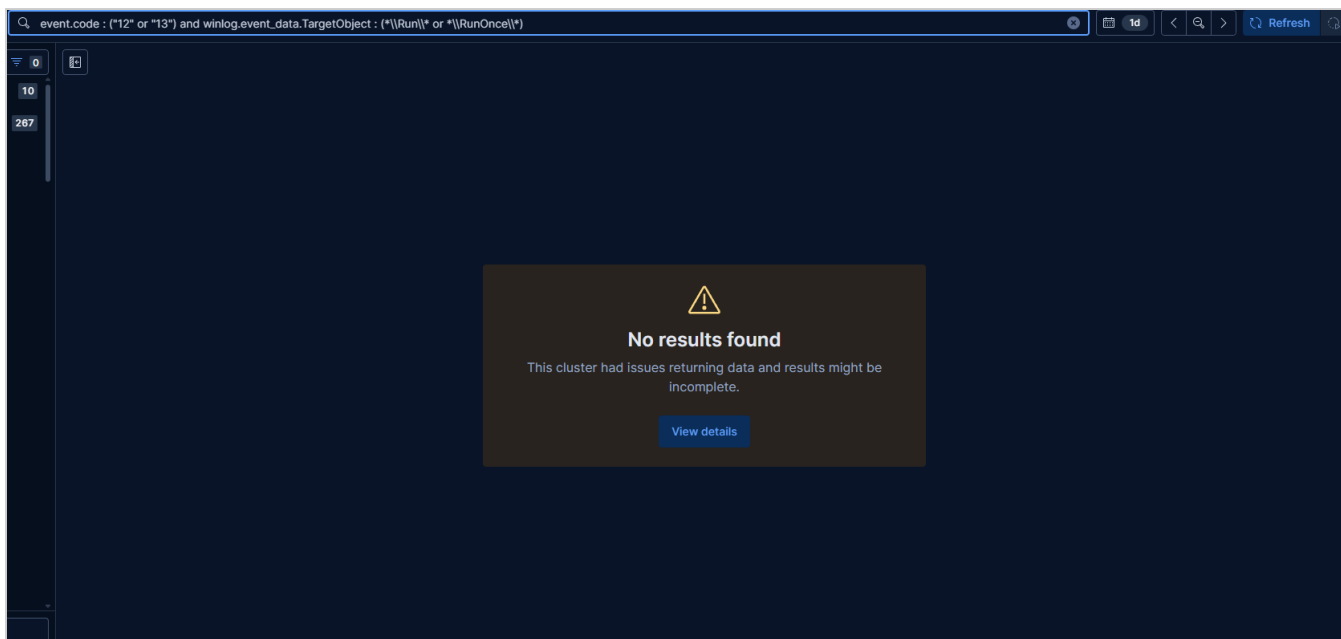


Figure 9 — Registry Run/RunOnce (Sysmon EID 12/13): no results. The `sc.exe` / `reg.exe` hits returned by the broader persistence query are shown in the appendix.

- [F] Registry Run / RunOnce writes (Sysmon EID 12/13) returned **no results** for this host and window.
- [F] The `sc.exe` and `reg.exe` events surfaced by the persistence query (tagged T1543.003 / T1012 by the Sysmon config) are all parented by `svchost.exe -k netsvcs -s Schedule` and run as SYSTEM/LOCAL SERVICE: `sc start wuauserv`, `sc start w32time`, `reg query ...\\SoftwareInventoryLogging`.
- [I] These are **benign Windows scheduled-maintenance** actions (Windows Update, time service, inventory logging), **not** attacker persistence — none is a child of `a3.exe`. No persistence attributable to the implant was found. The MITRE `RuleName` tags on these events are configuration labels, not analytic findings.

5.5 Lab ground truth (not investigative evidence)

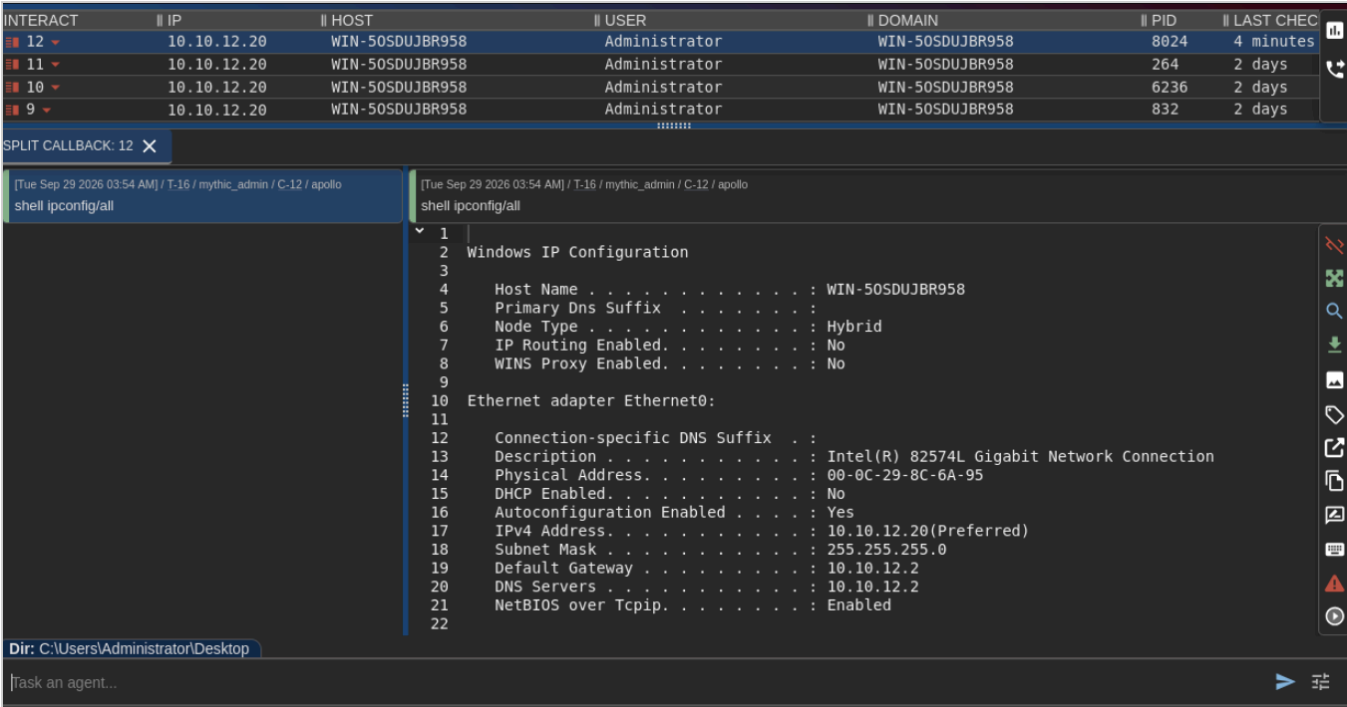


Figure 10 — C2 operator console (examiner-operated, ground truth): callbacks from 10.10.12.20 / WIN-50SDUJBR958 / Administrator (live callback PID 8024, matching the memory/EID-1 findings), and a `shell ipconfig/all` task via the `apollo` agent.

This screenshot comes from the examiner-operated C2 server. It would not be available in a real investigation and is not used to support any finding; it only confirms the pipeline's findings match the known ground truth of the lab exercise. The console shows several callbacks (a current one plus older ones from the 27/09 runs), consistent with the implant having been executed on the host more than once.

6. MITRE ATT&CK Mapping

Tactic	Technique	Basis
Initial Access	T1566.001 Spearphishing Attachment	[F]/[I] <code>a3.exe</code> extracted by WinRAR (archive delivery); email source not collected
Execution	T1204.002 User Execution: Malicious File	[F] WinRAR → <code>a3.exe</code> , then <code>explorer.exe</code> → <code>a3.exe</code>
Execution	T1059.003 Windows Command Shell	[F] <code>a3.exe</code> (PID 8024) → <code>cmd.exe /S /c ipconfig/all</code>
Discovery	T1016 System Network Configuration Discovery	[F] <code>ipconfig /all</code>
Defense Evasion	T1055 / T1620 Injection / Reflective Loading	[I] inferred from unbacked RWX memory
Command and Control	T1071.001 Web Protocols (HTTPS)	[F] <code>a3.exe</code> → 192.168.1.7:443

Notes: (1) the Sysmon `RuleName` labelling network-connect events as T1036 is a configuration label, not an analytic finding. (2) T1543.003 / T1012 `RuleName` tags appeared on **benign Windows maintenance** (`svchost -s Schedule, SYSTEM`) during the persistence check (Section 5.4) — they are not attributed to the implant.

7. Indicators of Compromise (IOCs)

Network

Type	Value	Context
IPv4 : port	192.168.1.7:443	C2 (lab) — <code>a3.exe</code>
Victim port	49923	Ephemeral port of this incident's C2 session

Host

Type	Value	Context
File path	C:\Users\Administrator\Desktop\a3.exe	Implant (.NET), delivered via WinRAR
SHA256	4d0ad28aea96e37a3a548c210bb7fb049ddd22a1d2aa61954ba2696f9a877b48	<code>a3.exe</code> binary (Sysmon EID 15)
SHA1	6e731fe2351c516673b123aebb72c0acf0118d5a	<code>a3.exe</code> binary
MD5	4cc8c5e06240f380709b690ff0de8f4b	<code>a3.exe</code> binary
IMPHASH	f34d5f2d4577ed6d9ceec516c1f5a744	<code>a3.exe</code> import hash (hunt for related samples)
Process	<code>a3.exe</code> (PID 8024, parent <code>explorer.exe</code>)	Implant
Memory	Unbacked RWX private regions in the <code>a3.exe</code> process	Injection indicator

The hashes above are the genuine binary's, captured by Sysmon at write time (Section 5.3) — usable IOCs, corrected from earlier drafts. IMPHASH is useful for finding related samples built the same way.

8. Conclusions

Against the investigative questions (Section 2.2):

1. **Yes** — [F] an established C2 session existed (`a3.exe` → 192.168.1.7:443), caught independently by Suricata (network) and the AI-SIEM (file).
2. `a3.exe`, delivered as a **WinRAR-extracted archive** to the Administrator Desktop (07:48:58 UTC) and executed by the user — the delivery vector is now **evidenced** (Section 5.2), not merely inferred.

3. **Yes** — [F] `a3.exe` (PID 8024) spawned `cmd /S /c ipconfig/all` for reconnaissance (07:54:20 UTC).
4. **No implant persistence found** — [F] `Run/RunOnce` empty; the flagged `sc.exe/reg.exe` activity is benign Windows maintenance (Section 5.4).
5. IOCs — including the genuine binary hashes and IMPHASH — are listed in Section 7.

Confidence: **High** for the C2, its process ownership, delivery and recon (each backed by ≥ 1 direct log source, several by two). **Medium** for the injection interpretation, which rests on memory-region attributes without binary reverse engineering.

9. Recommendations / Remediation

Containment - Isolate 10.10.12.20; preserve E3 and the disk. - Terminate/quarantine `a3.exe`; block 192.168.1.7:443 at pfSense.

Eradication / recovery - Re-image the endpoint; treat host Administrator credentials as compromised and rotate them. - Block the binary hashes (Section 7); optionally carve `a3.exe` from E3 (`windows.dmpfiles`) to confirm the Sysmon-captured hash. - Recover and inspect the source WinRAR archive and its delivery (email/URL) — the next pivot for initial access.

Detection hardening - Alert on unbacked RWX allocations in .NET processes and on `explorer` → `unknown.exe` → `cmd` → `ipconfig` chains. - For encrypted C2, use JA3/JA4, TLS SNI/cert and beacon-timing analytics rather than payload signatures. - Fix the AI-SIEM 0-byte false negative: do not let size features dominate; add parent-process and network context.

Process - Hash network evidence (E1, E2) at creation, not only at analysis time. - Run the Velociraptor server on the dedicated host (10.10.12.22), never on a monitored endpoint, so evidence is not staged on the compromised system. - Enforce and record NTP synchronisation; set Kibana `dateFormat:tz = UTC` and normalise all timelines to UTC.

10. Appendices

Full tool outputs are provided as files under `appendix/`:

The Kibana figures (7, 8, 6) are the Discover **summary** view, which does not display the decisive fields. Those fields — the creating process (`WinRAR.exe`), the binary hashes/IMPHASH, and the parent PID / command line — are taken from the **full JSON documents** exported from Kibana and stored in the appendix (A8–A10); the figures show the same events for context.

File	Content
appendix/A1_windows_info.txt	windows.info (this incident: SystemTime 2026-09-29 07:51:03 UTC)
appendix/A2_windows_netscan.txt	windows.netscan
appendix/A3_windows_pstree_reduced.txt	windows.pstree
appendix/A4_windows_cmdline_6236.txt	windows.cmdline
appendix/A5_windows_dlllist_6236.txt	windows.dlllist
appendix/A6_windows_filescaan_a3.txt	windows.filescaan matches for a3.exe
appendix/A7_windows_malfind_6236_regions.txt	windows.malfind region table
appendix/A8_eid11_filecreate_a3.json	Sysmon EID 11 (full JSON) — process.executable = WinRAR.exe, file.path = ...a3.exe
appendix/A9_eid15_streamhash_a3.json	Sysmon EID 15 (full JSON) — file.hash.sha256/sha1/md5, file.pe.imp hash
appendix/A10_eid1_execution_a3.json	Sysmon EID 1 (full JSON) — winlog.event_data.CommandLine, ParentProcessId 8024, IntegrityLevel High
appendix/A11_persistence_check.csv	Persistence check (full export) — benign Windows maintenance only

References - NIST SP 800-86 — Guide to Integrating Forensic Techniques into Incident Response. - NIST SP 800-61 Rev. 3 — Incident Response Recommendations. - ISO/IEC 27037:2012 — Identification, collection, acquisition and preservation of digital evidence. - ISO/IEC 27041:2015 — Assuring suitability and adequacy of investigation methods. - ISO/IEC 27042:2015 — Analysis and interpretation of digital evidence. - MITRE ATT&CK — <https://attack.mitre.org/> - Volatility 3 — <https://volatility3.readthedocs.io/>